

IT24100327 Perera B.P.N

Lab 10 - Probability & Statistics

R Script

```
1 # Perera B.P.N - Lab 10
2 # IT24100327
3
4 # H0: All snack types are chosen with equal probability.
5 # H1: At least one snack type is chosen with a different probability.
6
7 observed <- c(120, 95, 85, 100)
8
9
10 prob <- c(0.25, 0.25, 0.25, 0.25)
11
12 test_result <- chisq.test(x = observed, p = prob)
13
14 print(test_result)
15
16 # At  $\alpha = 0.05$ :
17
18 # p-value = 0.089 > 0.05 → Fail to reject H0
19
20 # Conclusion:
21 # There is no significant difference among the snack type choices.
22 # Customers appear to choose snacks equally.
```

Output

Chi-squared test for given probabilities

data: observed

X-squared = 6.5, df = 3, p-value = 0.08966